

Event-Driven Earnings Drift Model

(Long vs Short)

This technical study documents the methodology, mathematics, and operational logic of the Event-Driven Earnings Drift Model implemented on ports 8080 (Long-Only) and 8000 (Short-Selling).

1. Executive Summary & Strategy Core

The core methodology is built on the **Earnings Drift Anomaly** positioning. Instead of predicting the direction or surprise of a company's upcoming earnings report, the model positions around the **Catalyst Announcement Date (Result Date)**.

By analyzing a stock's historical price behavior around its past earnings announcements, the engine optimizes entry and exit offsets to capture:

- **The Pre-Result Anticipation Drift** ("Buy the Rumour" run-up or "Sell the Rumour" profit booking).
- **The Post-Result Digest Drift / PEAD** (Post-Earnings Announcement Drift as the market digests the surprise).

2. Leg Mechanics & Mathematical Formulation

Every event-driven trade is executed as a **two-leg transaction** matching specific trading-day offsets relative to the Result Announcement Date (T).

A. Long Strategy (Buy Before, Sell After)

Used when a stock exhibits consistent upward momentum leading into or immediately following results.

- **Entry (Buy)**: Executed at the Closing Price on trading day $T - N$.
- **Exit (Sell)**: Executed at the Closing Price on trading day $T + M$.

Math Formulation:

$$\text{Long Return (\%)} = [(\text{Price}_{\text{exit}} / \text{Price}_{\text{entry}}) - 1 - \text{Cost}] * 100$$

Where Cost represents transaction costs / slippage fraction (default: 0.10% or 0.0010 round-trip).

B. Short Strategy (Sell Before, Buy After)

Used when a stock exhibits consistent downward momentum or pre-result profit booking.

- **Entry (Short Sell):** Executed at the Closing Price on trading day $T - N$.
- **Exit (Cover Buy):** Executed at the Closing Price on trading day $T + M$.

Math Formulation:

$$\text{Short Return (\%)} = [(\text{Price}_{\text{entry}} - \text{Price}_{\text{exit}}) / \text{Price}_{\text{entry}} - \text{Cost}] * 100$$

3. How the Engine Predicts the Candle (Grid Optimization)

The model determines the optimal buy and sell offsets (N and M) using **ex-ante historical grid backtesting** over the past quarters:

```

[ ] | Historical Event Dates |
[ ] | [For each (N, M) combination] Long: N,M ∈ [1, 8] |
Short: N ∈ [1, 10], M ∈ [1, 15] ▼ [ ] | Simulate Trade
Returns | [ ] | ▼ [ ] |
Calculate Metrics per Stock: | | - Average Return (%) | | - Win Rate (%) |
[ ] | ▼ [ ] | Pick optimal
parameters. | | (Max Win Rate > Max Return) | [ ]
  
```

1. Parameters Grid Search:

- **Long-Only Grid:** Entry Offset (N) from 1 to 8 days, Exit Offset (M) from 1 to 8 days (**8x8 grid**).
- **Short-Selling Grid:** Entry Offset (N) from 1 to 8 days, Exit Offset (M) from 1 to 8 days (**8x8 grid**).

2. **Backtesting Matrix:** The engine evaluates a symmetric 64 combinations (8x8) for both Long and Short strategies across historical earnings result dates to prevent overfitting.

3. **Objective Optimization Function:** The optimal offset combination is chosen by ranking combinations first by **Win Rate (%)**, and then by **Average Return (%)**.

4. Port 8080 (Long-Only) vs Port 8000 (Short-Selling) Comparison

DIMENSION	PORT 8080 (LONG-ONLY)	PORT 8000 (SHORT-SELLING)
Operational Logic	Buy first at T-N, sell later at T+M.	Sell first at T-N, buy back (cover) at T+M.
Market View	Bullish anticipation / post-result reaction.	Bearish anticipation / negative earnings drift.
Profit Trigger	Exit Price > Entry Price	Exit Price < Entry Price
Risk Profile	Maximum loss is limited to 100% of capital.	Maximum loss is theoretically unlimited.
Capital Allocation	Cash used to purchase asset shares.	Cash locked as margin to support short position.

5. Summary Matrix (Losing Trades Decision Logic)

Our backtest highlights that having a combined strategy selection model is optimal. Even when trades result in a loss, the decision matrix selects the leg that minimizes losses. For example:

- **NTPC (Q2):** Choosing SHORT resulted in a **-0.45%** loss, avoiding the **-1.18%** loss from the LONG strategy.
- **SHRIRAMFIN (Q2):** Choosing LONG resulted in a **-1.49%** loss, avoiding the massive **-12.32%** loss from the SHORT strategy.